The cell–cell junction:

myosin as a density on a belt, through a division

`prototype/ecm/junction_ops.py` · `medioapical_ops.py` · `test_01{,b,c}*.py` · `log/okuda_ECM/01, 01b, 01c` · 10 August 2026

1. Mechanism

In a vertex model every cell–cell contact pulls with the same strength. Real contractility is not uniform: myosin accumulates on some junctions and not others, and where it accumulates the junction pulls harder and shortens, which is how cells exchange neighbours.

A junction is a surface, and it enters the energy as a line. The lateral interface between two cells is of order $\ell \times h$, ℓ its apical extent and h the cell's height. This epithelium is one triangulated shell — a cell is a face and its "volume" is the wedge from the tissue centre — so there is no h, and the junction enters as a line of length ℓ_e with the third dimension absorbed into the line tension Λ. That is the right geometry for the term it multiplies rather than a compromise: the contractile actomyosin of an adherens junction is a *belt* at the apical rim of that interface, a cable of length ℓ_e; what is spread over the lateral *area* is cadherin adhesion, which this energy does not carry.

The consequence is what the symbol means. $\Lambda \sum_e m_e \ell_e$ is a tension times a length, so m_e is a **density** — myosin per unit length of belt — and the *amount* on a junction is $N_e = m_e \ell_e$. Everything below follows from that identification, and the model's own inheritance rule is only correct if it is true: when `divide_3d` cuts a contact (a, b) into (a, n) and (n, b), *nothing physical has happened* — the same two cells still touch across the same interface — and copying the parent's m onto both halves conserves the amount only because m is intensive. Had m been an amount, the identical line of code would have doubled the tissue's myosin at every division and nothing would have failed.

Two pools, because the biology has two. An epithelial cell carries a **medioapical** meshwork spread over its apical face and a **junctional** belt at the rim, coupled by outward flow, and germband extension uses both [3]. The first is areal, the second linear. So "should myosin scale with area or with length" has the answer *both*, and the repair for the defect above is not a better drive but a conserved amount with a derived density. In a dividing cell, almost all of the myosin goes to the **cytokinetic ring**, which constricts exactly at the nascent interface; in epithelia the new adherens junction is built out of it, with the neighbouring cell contributing [5, 6].

2. Equations and symbols

The energy the myosin multiplies, unchanged from the stock 3D vertex model except for m_e:

$$E = \sum_f \Big[K_A(A_f - A_f^0)^2 + K_P(P_f - P_f^0)^2 + \tfrac{1}{2}\Gamma P_f^2 + K_V(v_f - v_f^0)^2\Big] + \Lambda \sum_e m_e \ell_e + K_R \sum_i (|\mathbf{x}_i| - R_0)^2. \quad (1)$$

With $m_e \equiv 1$ this is the stock Okuda energy [1, 2] and every prior run is bit-identical.

The two pools. ρ_f is areal and lives on the cell; n_e is linear and lives on the junction; the flux between them is per unit length of belt, because there is ℓ_e of belt to receive it:

$$M_f = \rho_f A_f, \qquad \frac{\mathrm{d}M_f}{\mathrm{d}t} = k_{\mathrm{on}} A_f - \frac{M_f}{\tau_{\mathrm{med}}} - \sum_{e \in \partial f} J_{f\to e}, \qquad J_{f\to e} = k_{\mathrm{ex}} \rho_f \ell_e \big(1 + \beta_T (T_e/\langle T\rangle - 1)\big), \quad (2)$$

$$N_e = n_e \ell_e, \qquad \frac{\mathrm{d}N_e}{\mathrm{d}t} = \sum_{f \ni e} J_{f\to e} - \frac{N_e}{\tau_{\mathrm{jun}}}, \qquad m_e = a\, n_e/\langle n\rangle. \quad (3)$$

The setpoint is now intensive, which is the whole point: the belt's supply is per unit length, so

$$n_e^* = \tau_{\mathrm{jun}} k_{\mathrm{ex}} (\rho_f + \rho_g) \quad (4)$$

— two-sided, because a junction is fed by both cells it separates — carries no ℓ_e, and half a junction relaxes toward what the whole junction was relaxing toward. The one-pool law does not: its target is $m_e^{ss} = a\ell_e/\langle \ell \rangle$, and ℓ_e is extensive, so a cut halves it.

the drive of a feedback must be intensive.

Tension is (cutting a cable in two does not halve the tension in it); strain and strain rate are; length is not.

Figure 1: Two cells share an edge, and that edge *is* the junction — it belongs to neither, which is why its state is keyed by the pair of vertices rather than stored on a cell. Myosin sits on the shared edge and multiplies its tension. Right: the feedback, which is the only thing here the original model cannot already do.

The cytokinetic ring seeds every interface with *both* endpoints new — exactly the case the inheritance rule excludes — and *debits* the two daughters' cortical pools, so it moves myosin instead of creating it:

$$n_e = \texttt{ring} \times n^*, \qquad \rho_{f,g} \text{ reduced by the same amount.} \tag{5}$$

One prediction arrives unbidden. The steady state of Eq. (2) is $\rho^* = k_{\text{on}}/(1/\tau_{\text{med}} + k_{\text{ex}} P_f/A_f)$: a cell with more perimeter per unit area drains faster and holds less medioapical myosin. P_f/A_f is fixed by shape alone, so the model says — with nothing added to make it say so — that as a tissue divides into smaller cells, myosin shifts off the cortex and onto the belts. That is gate J8.

symbol	is	of what
m_e	myosin on junction e	a DENSITY, dimensionless multiplier on Λ, mean 1
n_e, N_e	line density, amount	$N_e = n_e \ell_e$; what is stored is n_e
ρ_f, M_f	areal density, amount	$M_f = \rho_f A_f$, on the cell's apical face
k_{on}	assembly rate	myosin assembled per unit *apical area* per frame
k_{ex}	export rate	drawn from the cortex per unit *length* of belt per frame
$\tau_{\text{med}}, \tau_{\text{jun}}$	turnover times	in FRAMES; see the caveat at the end of §4
β_T	tension bias on the flux	0 in the reported run: pure transport, deliberately
`ring`	the ring's deposit	in units of n^*, so it cannot drift with the tissue's mean
a	global activity	multiplies every junction equally
Λ, Γ, K_P	line tension, contractility, perimeter stiffness	read OFF `shape_energy_3d`, not restated — a tension computed from different constants is the tension of a different tissue

3. The Plexus2 container

One structural claim. A junction is a *relation between two vertices*, not a body: it belongs to neither cell, so its state is keyed by the vertex pair and not stored on a face. That is what makes the inheritance rules local and what makes the sync operator below possible at all.

set	**kind**	**one element is**	**state it carries**
`cell`	node, depth 1	one epithelial cell	A, P, V and their targets, age, `ndiv`, ρ_f
`vertex`	node, depth 1	a corner of the apical mesh	`pos`
`junction`	**edge** $\subseteq$ `vertex`^{2}	a cell–cell contact	n_e, ℓ_e, m_e (written to `m["myo"]`)

operator	kind	acts on	mechanism	its gate
`medioapical_myosin`	Lateral	`cell`	Eq. (2); emits $J_{f\to e}$	J8
`junction_myosin [two_pool]`	**Structural**	`junction`	Eq. (3); integrates N_e, writes m_e	J1–J4
`cytokinetic_ring`	Structural	`junction`	Eq. (5); seeds newborns, debits the daughters	J5–J7
`junction_myosin_sync`	**Rewire**	`junction`	re-keys m_e, N_e by vertex pair after the topology operators	J9
`reconnect_t1_3d`, `divide_3d`	Rewire, Divide	`junction`, `cell`	neighbour exchange and cytokinesis [1]	J10

Three placements carry the argument. `two_pool` is registered as a `model=` of the existing contract and not an `implementation=`: its parameters $\{\tau_{\rm jun}, a\}$ are disjoint from the default's $\{a, \tau, \beta,$ `keyed_on`$\}$, so there is no common operating point at which to call them two ways of computing one thing — per plexus2, swapping a model *is* an experiment. They share the contract's kind and write the same channel, so `shape_energy_3d` cannot tell them apart and the comparison is fair. The medioapical operator is scheduled *before* the belt's, so a frame's flux is the flux that frame's cells produced. And the sync runs *after* the topology operators:

```
... -> medioapical_myosin -> junction_myosin -> shape_energy_3d
    -> reconnect_t1_3d -> divide_3d -> cytokinetic_ring -> junction_myosin_sync -> ...
```

The sync operator is why any of this is measurable. `junction_myosin` writes `m["myo"]` sized to the half-edge buffer at its own slot; `reconnect_t1_3d` rewires those arrays and `divide_3d` lengthens them later in the same frame, and `topo_snapshot_3d` then recorded the frame's edges beside the *previous* topology's myosin. Every division frame — which is precisely where this note's measurement lives — was one of the broken ones (J9).

4. Gates

A gate is a number with a threshold decided *before* the run and a stated consequence if it fails. Two builds of the same 401-frame tissue differ only in the myosin model, and within each, junctions a division *cut* are compared against *intact* ones over the same interval. The control is not optional: two frames of ordinary dynamics move an untouched junction too, and without it every division looks like a leak.

	gate	threshold	measured
the amount survives a re-meshing (`01b`, 13,083 / 12,320 cuts)			
J1	$(N_1{+}N_2)/N_{\rm parent}$ over $(\ell_1{+}\ell_2)/\ell_{\rm parent}$, cut	$\le$ the intact control	one-pool 1.0068, two-pool **1.0043**
J2	the same ratio on intact junctions (the control)	—	1.0092 / 1.0093
J3	setpoint of a half over its parent's	$= 1$; length-keyed gives $\frac{1}{2}$	one-pool **0.514**, by design
J4	m twenty frames after the cut, over intact	> 0.90	one-pool **0.797**, two-pool **0.939**
the newborn junction (`01c`)			
J5	daughter–daughter interface, $m/\langle m\rangle$ at birth	within 0.10 of 1	absolute **0.628**; ring= 1 **0.920**; ring= 3 2.483
J6	drift of that value across the run	< 0.15	absolute **0.33**; ring= 1 **0.12**
J7	the deposit relaxes on the belt's own timescale	e-fold within 20% of $\tau_{\rm jun} = 20$	**∼19** frames (2.48 → 1.22 over 30)
the prediction nothing was added to make (`01b`)			
J8	medioapical share falls as $\langle P_f/A_f\rangle$ rises	monotone, both	share 0.214 → **0.096**, P/A 3.38 → **5.19**
the bookkeeping (`01`, the nominal)			
J9	myosin array length equals the edge array's	0 of 200 short	**56** before the sync (by 6–1356); **0** after
J10	the sync perturbs nothing: positions, divisions, T1s	bit-identical	**bit-identical**
J11	T1 per cell per frame: does the law change the observable	any separation	0.00221 vs **0.00373** (1.69×)
the scale the note is quoted in (`0u_units`)			
U1	one frame, from 4.93 doublings at a 12–24 h cycle	$[8.8, 17.7]$ min	**10 min**
U2	$\tau_{\rm jun} = 20$ frames — maturation, NOT myosin turnover (1–2 min)	$[30, 600]$ min	**200 min**
U3	cell diameter (the length CALIBRATION)	$[5, 15]$ um	**8.5 um**
U4	junction length at the last frame	$[2, 15]$ um	**4.6 um**
U5	time between neighbour exchanges, per cell	$[20, 200]$ h	**44.68 h**

The run declares its scale, and these numbers are what it makes of it. Three base scales under `general.units:` and everything else derived (`plexus/units.py`): 1 box unit = 1172 μm, because a cell is 8.5 μm and the tissue is solved at 10 μm per tissue unit; 1 frame = 600 s, because 200 → 6,076 cells is 4.93 doublings and a 12–24 h cycle puts the run at 2.5–4.9 days; 1 force unit = 9162 nN, set so the stroma's `youngs: 15` is 100 Pa — a collagen I gel at 1–3 mg/ml. Only the force scale is *chosen*, and it is chosen on the softest and best-measured body rather than on the sheet under dispute. Without this block the loader says the run is dimensionless and no result from it may carry a unit, which was the state of every run in this folder until now.

Two conventions make these mean what they say. A cut is measured against an intact control on the same frames, because two frames of dynamics move an untouched junction by 0.9% and a raw ratio of 1.007 would otherwise read as a leak. And **a newborn's value is read against the tissue's own mean, binned by time**: `myo_new` was an absolute line density in a model whose mean line density runs 1.07 → 1.97 → 1.48 over 401 frames, so what a new junction started with was decided by the frame index — the run-median hides that, and J6 is the gate that does not.

The residual in J4 is not a leftover artifact. Writing $N_e = n_e \ell_e$ in Eq. (3) gives $\dot{n}_e = k_{\text{ex}}\rho - n_e/\tau_{\text{jun}} - n_e\,\mathrm{d}\ln \ell_e/\mathrm{d}t$: a junction whose length is changing dilutes or concentrates its own belt. Freshly cut halves relax and lengthen, so they dilute. That term is advection of a conserved species along a moving cable, which is physics; the 20% it replaces was arithmetic. The same term is why J5 sits at 0.92 and not 1.00 — mature junctions are not *at* n^*, because the tissue's mean junction length falls $0.75 \to 0.46$ as cells divide and a shortening belt concentrates above its own steady state.

What the gates do not license. The operating point: k_{on} was set to $1/\tau_{\text{med}} + k_{\text{ex}}\langle P/A\rangle_0 = 0.219$ so the cortical density *starts* at its own steady state, and $\tau_{\text{med}} = \tau_{\text{jun}} = 20$ frames with $k_{\text{ex}} = 0.05$ are chosen numbers. $\beta_T = 0$ in the reported run — pure transport, deliberately, so a tension bias added later has to beat what geometry already does (J8). There is no pulsatility, which is the mechanism the medioapical pool is famous for: Munjal's pulses need advection-mediated positive feedback with dissociation as the negative arm, a third state variable and an advection term. And **one frame is still not anchored to a time**: the run's own growth fixes it at 9–18 minutes per frame, at which junctional myosin turnover (1–2 minutes) is a tenth of a frame and the belt is adiabatic. A 20-frame memory on a junction can legitimately be *maturation* — E-cadherin accumulation over tens of minutes to hours — so τ_{jun} should be renamed and re-sourced rather than quoted as turnover.

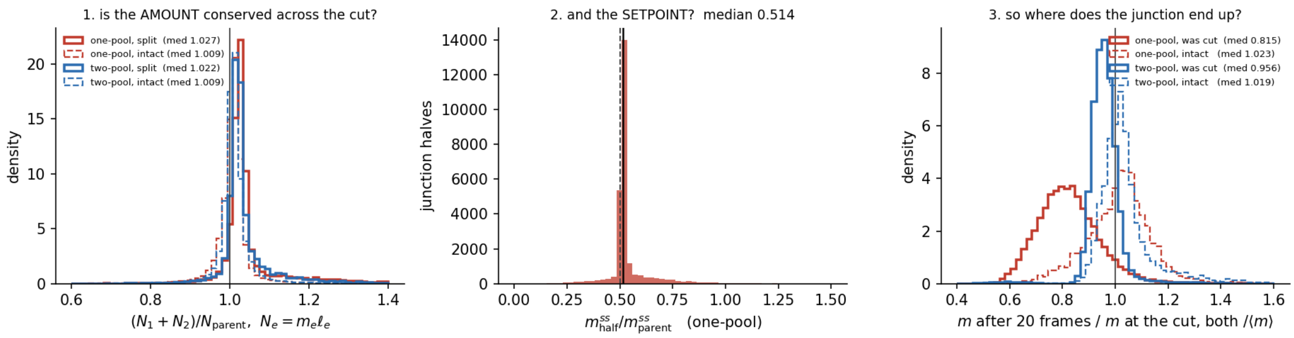

Figure 2: J1–J4. Left: the amount across a cut, both models, against the intact control — the offset from 1 is the 2% by which the two halves are longer than the parent they replace, not a leak. Middle: the one-pool setpoint of a half over its parent's, which is 1/2 because the setpoint is proportional to a length that was just halved (J3). Right: where the junction actually ends up twenty frames later (J4).

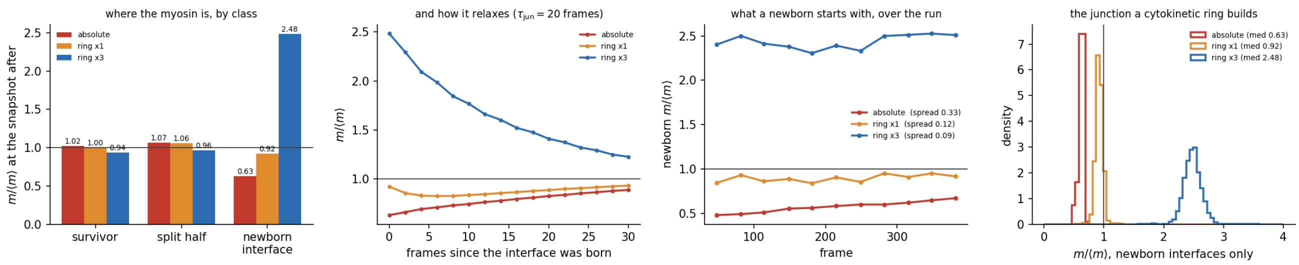

Figure 3: J5–J7. Left: myosin by junction class. Middle-left: the deposit relaxing — `ring`= 3 falls $2.48 \to 1.22$ over 30 frames toward the `ring`= 1 level, an e-folding of $\sim$19 frames against a τ_{jun} of 20 that nobody fitted: the ring needs no timescale of its own because the belt already has one. Middle-right: J6, what a newborn starts with, binned by frame — the absolute rule drifts $0.479 \to 0.670$, both ring runs are flat.

5. References

References

[1] Okuda, S., Inoue, Y., Eiraku, M., Sasai, Y., Adachi, T. (2013) Reversible network reconnection model for simulating large deformation in dynamic tissue morphogenesis. *Biomech. Model. Mechanobiol.* **12**(4):627–644.

[2] Okuda, S., Miura, T., Inoue, Y., Adachi, T., Eiraku, M. (2018) Combining Turing and 3D vertex models reproduces autonomous multicellular morphogenesis. *Sci. Rep.* **8**:2386.

[3] Munjal, A., Philippe, J.-M., Munro, E., Lecuit, T. (2015) A self-organized biomechanical network drives shape changes during tissue morphogenesis. *Nature* **524**:351–355.

[4] Fernandez-Gonzalez, R., Simoes, S. de M., Röper, J.-C., Eaton, S., Zallen, J.A. (2009) Myosin II dynamics are regulated by tension in intercalating cells. *Dev. Cell* **17**(5):736–743.

[5] Herszterg, S., Leibfried, A., Bosveld, F., Martin, C., Bellaiche, Y. (2013) Interplay between the dividing cell and its neighbors regulates adherens junction formation during cytokinesis in epithelial tissue. *Dev. Cell* **24**(3):256–270.

[6] Founounou, N., Loyer, N., Le Borgne, R. (2013) Septins regulate the contractility of the actomyosin ring to enable adherens junction remodeling during cytokinesis of epithelial cells. *Dev. Cell* **24**(3):242–255.